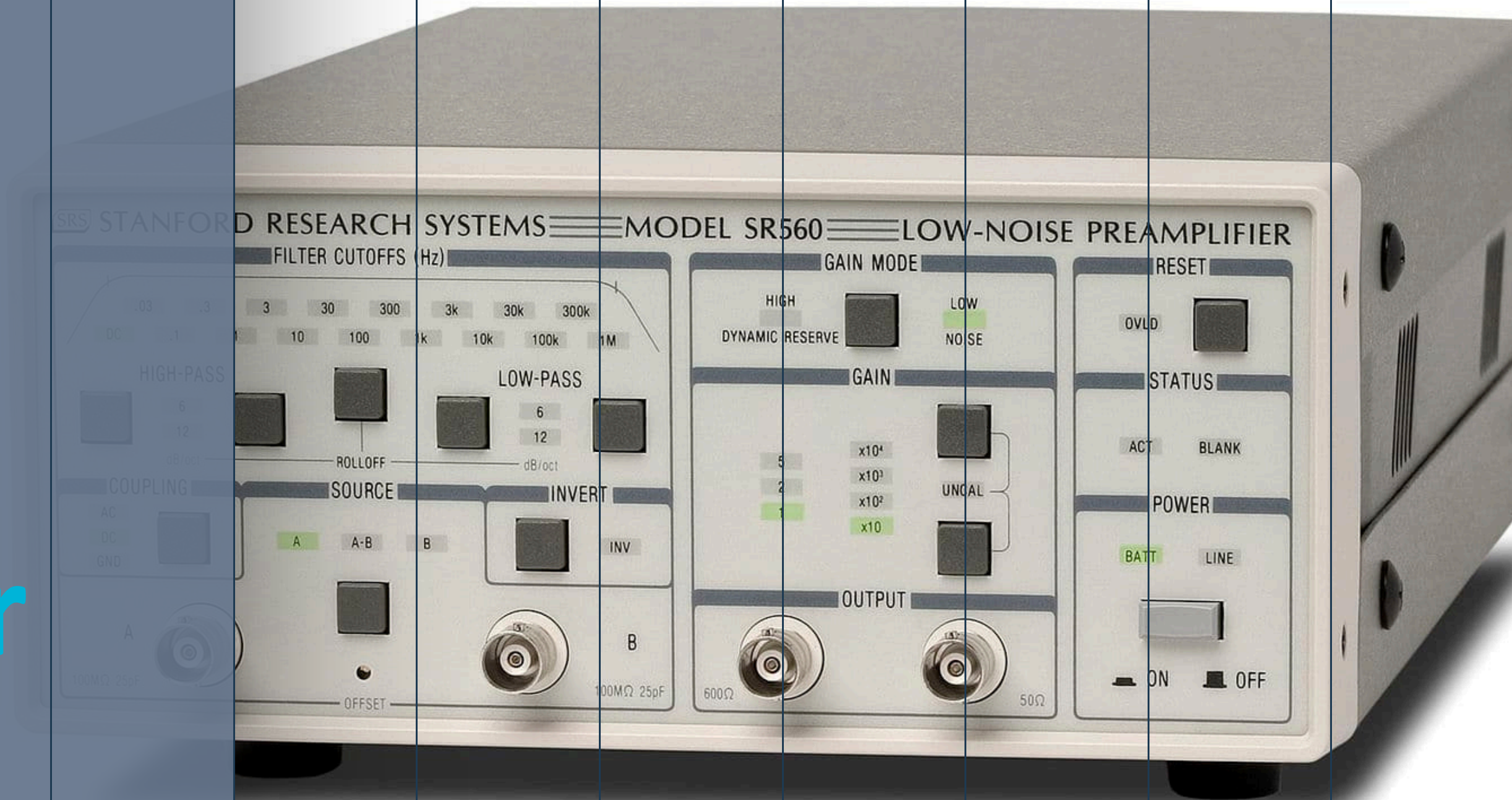


Active Noise Canceling Pre-Amplifier



The Interference Problem

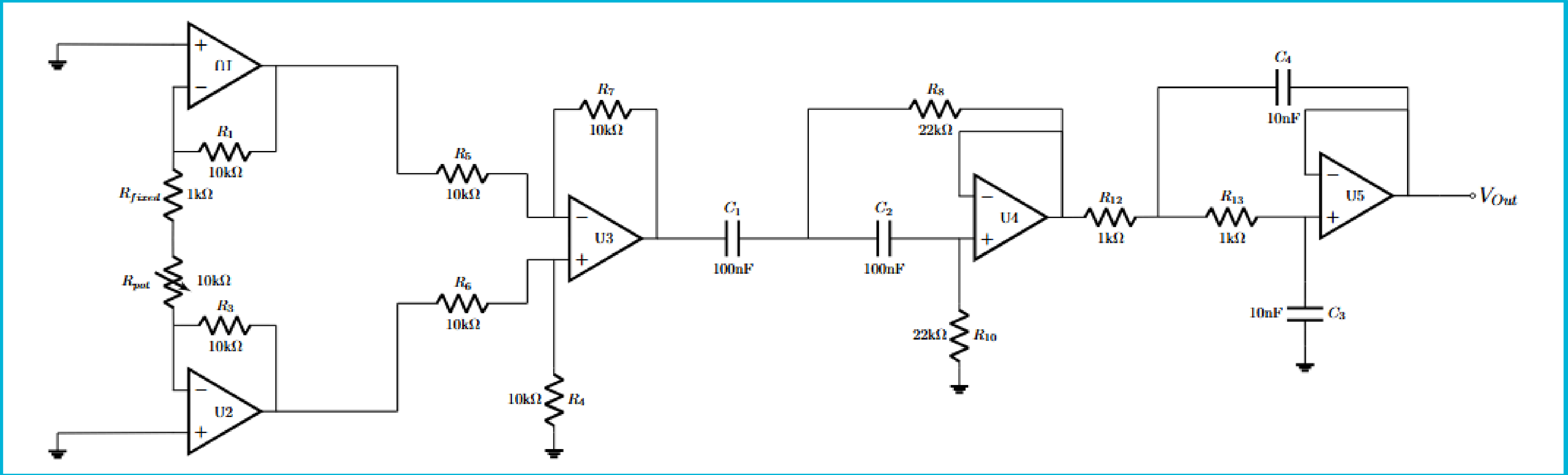
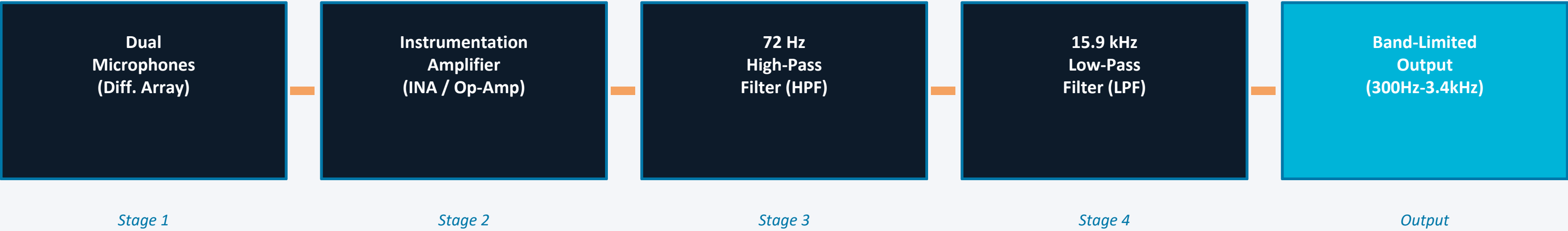
- Mains hum: 50 Hz fundamental + harmonics (100, 150, 200 Hz...)
- Human vocal band: 300 Hz - 3.4 kHz (ITU-T G.711)
- Interference amplitude can exceed signal by >40 dB in unshielded environments
- Single-ended microphone inputs couple noise as a common-mode signal

Why Analog Pre-Processing?

- ADC saturation: high-amplitude noise consumes dynamic range before digitization
- Digital DSP cannot recover clipped data — information is permanently lost
- Analog rejection must precede the ADC in the signal chain
- Differential input architecture rejects common-mode noise at the source

Objective: Reject 50 Hz interference while preserving 300 Hz - 3.4 kHz vocal content with <1 dB passband ripple.

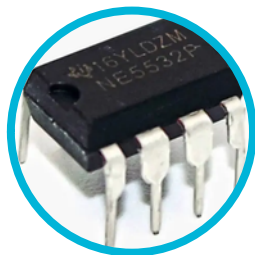
02 — System Architecture & Hardware Implementation





Passive Components

- E24 series resistors
- COG/NPO ceramic capacitors (+-5%, temp-stable)
- Standardized values minimize deviation from calculated f_c
- Sourced locally: Tronic.lk / Duino.lk
- Schottky diodes were used because their low forward voltage drop and fast switching characteristics
- Zener diodes were used to regulate and stabilize the output voltage

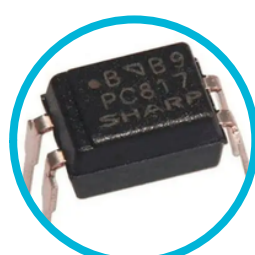


Op-Amp Selection

- Primary: NE5532 (low-noise, dual, audio-grade)
- Alternative: TL074 (quad, JFET input, low bias current)
- Both: $V_{os} < 5\text{ mV}$, $GBP \geq 10\text{ MHz}$
- Sourced locally: Tronic.lk / Duino.lk

FEATURES

- Small-signal bandwidth: 10MHz
- Output drive capability: 600Ω , $10V_{RMS}$
- Input noise voltage: $5\text{ nV}/\sqrt{\text{Hz}}$ (typical)
- DC voltage gain: 50000
- AC voltage gain: 2200 at 10kHz
- Power bandwidth: 140kHz
- Slew rate: $9\text{ V}/\mu\text{s}$
- Large supply voltage range: ± 3 to $\pm 20\text{ V}$
- Compensated for unity gain

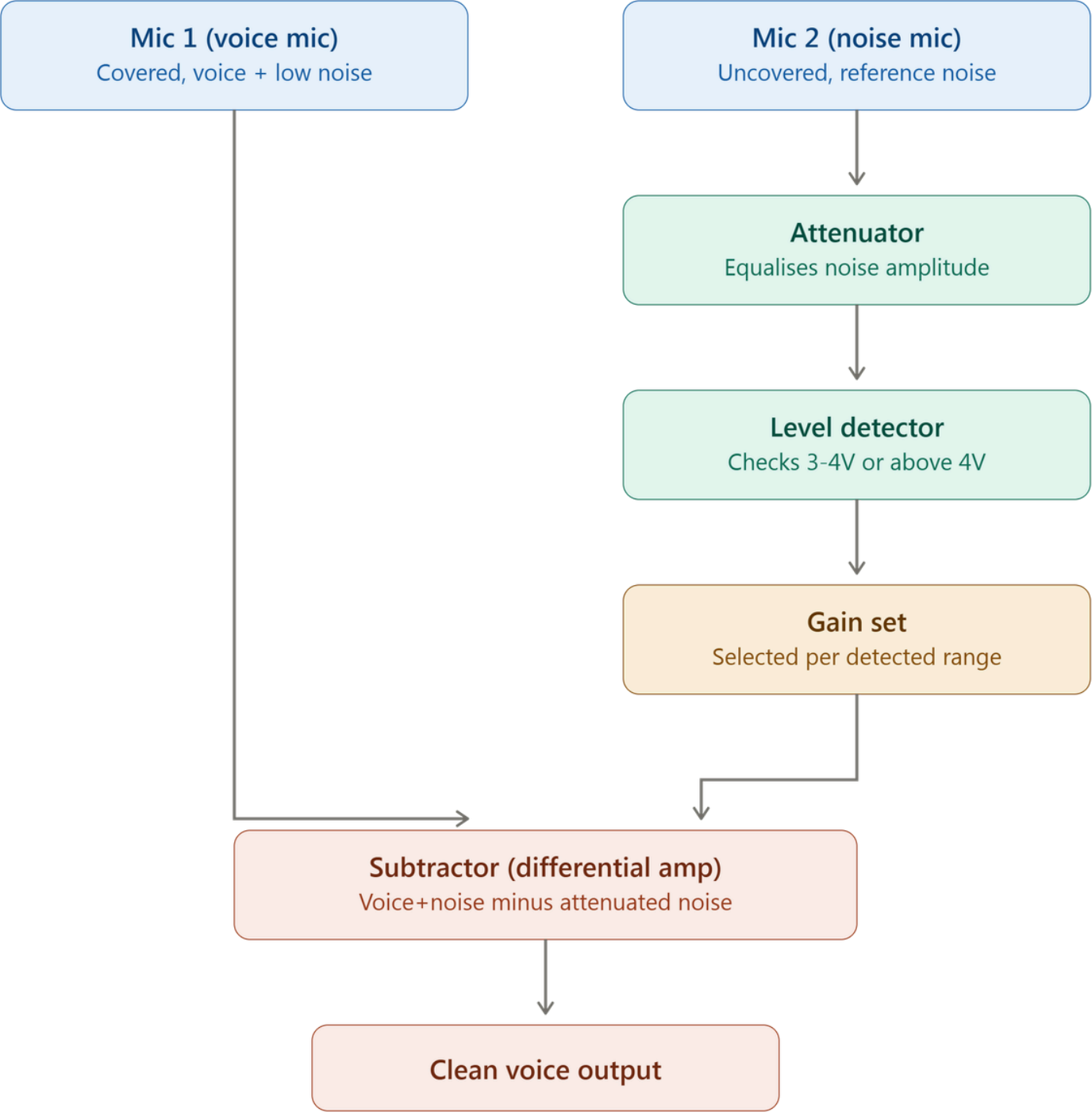


Optocoupler Selection

- An optocoupler was used to provide galvanic isolation between circuit sections, thereby reducing ground-loop hum, minimizing the coupling of external electrical noise into the low-level audio path, and protecting the sensitive preamplifier stage from interference and transients.

Parameter		Symbol	Rating	Unit
Input	*1 Forward current	I_F	50	mA
	*2 Peak forward current	I_{FM}	1	A
	Reverse voltage	V_R	6	V
	*1 Power dissipation	P	70	mW
Output	Collector-emitter voltage	V_{CEO}	35	V
	Emitter-collector voltage	V_{ECO}	6	V
	Collector current	I_C	50	mA
	*1 Collector power dissipation	P_C	150	mW
*1 Total power dissipation		P_{tot}	200	mW
*3 Isolation voltage		V_{iso}	5	kVrms
Operating temperature		T_{opr}	-30 to +100	°C
Storage temperature		T_{stg}	-55 to +125	°C
*4 Soldering temperature		T_{sol}	260	°C

04 — Working Principle of the Circuit



- **Capture two inputs** – Mic 1 (covered) picks up voice + low-level noise; Mic 2 (uncovered) picks up reference noise only.
- **Equalise noise amplitude** – Attenuate Mic 2's noise signal so its level matches the noise leaking into Mic 1.
- **Detect noise level** – Measure the equalised noise amplitude and classify it: 3–4 V, or greater than 4 V.
- **Set gain** – Based on the detected range, adjust the attenuation/injection level using the power supply.
- **Combine signals** – Feed Mic 1's voice+noise signal and the conditioned noise reference into a subtractor (differential amplifier).
- **Cancel noise** – Subtract the attenuated noise from the voice+noise signal to produce a clean voice output.

05 — Signal-to-Noise Ratio (SNR) Analysis

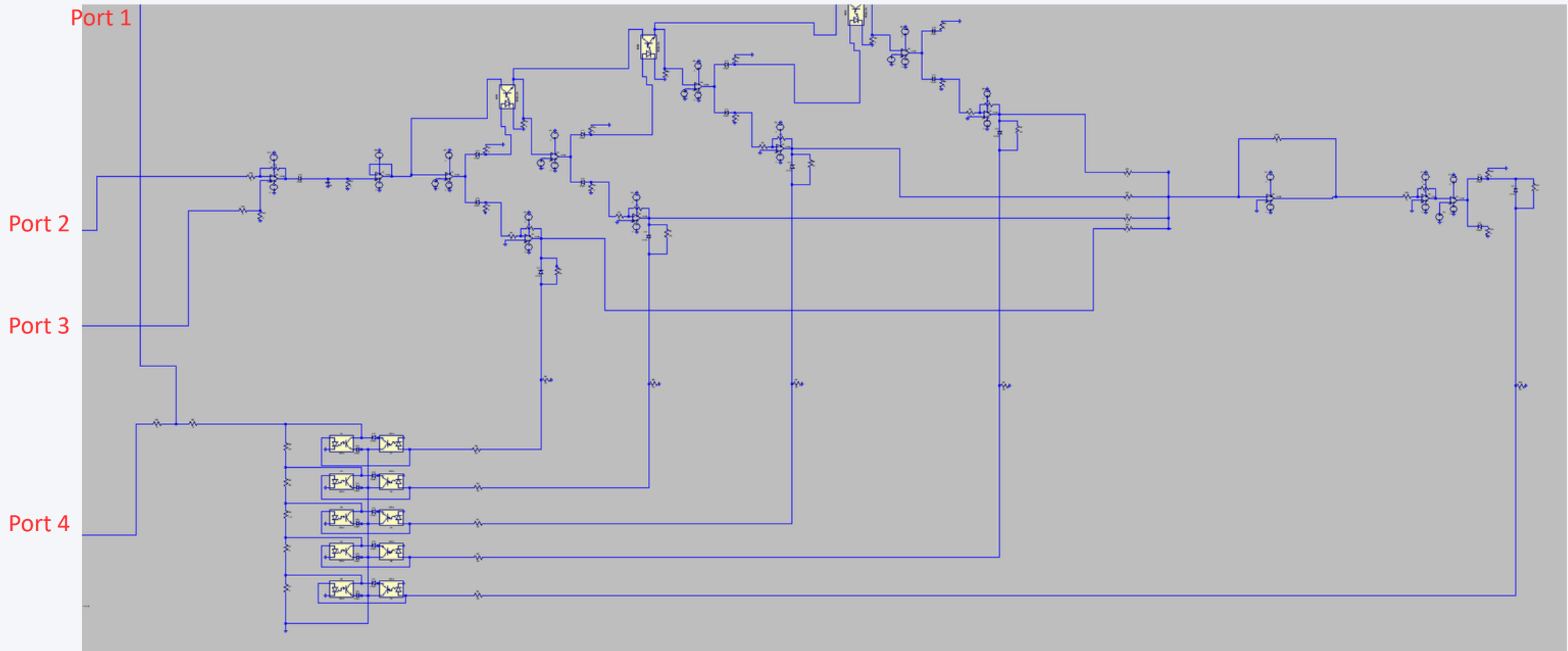
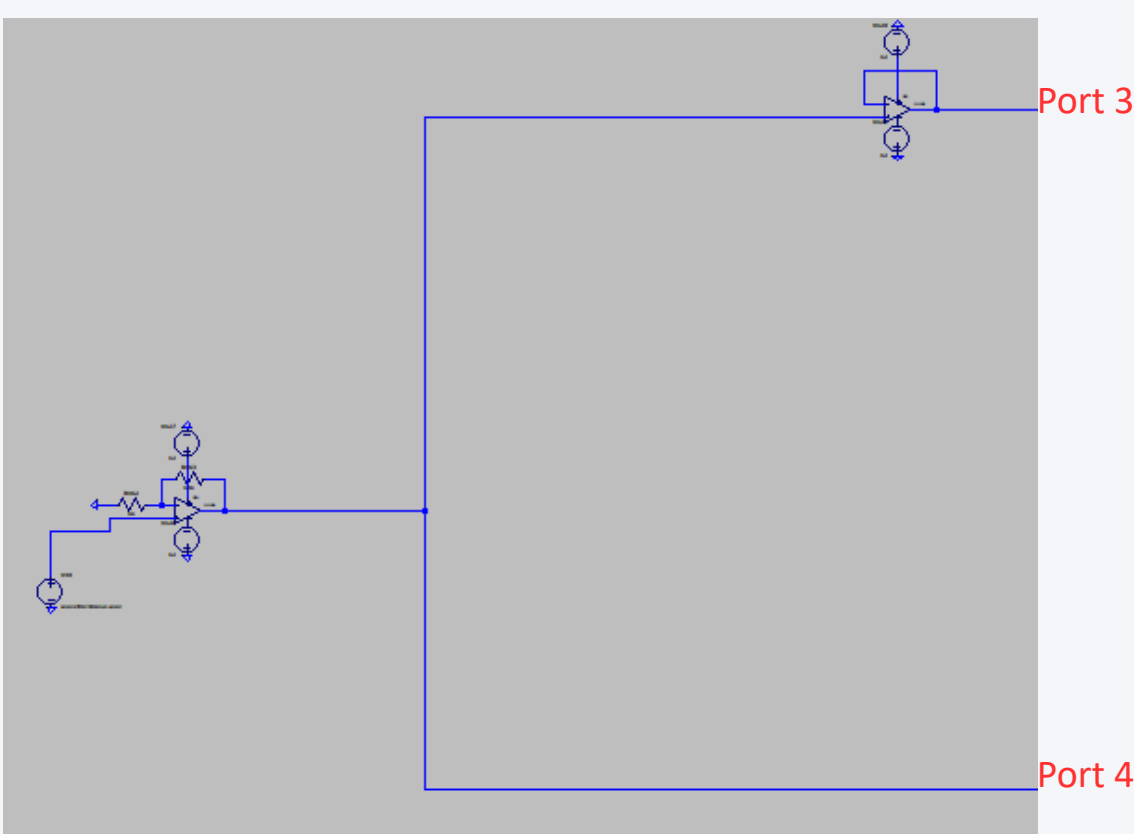
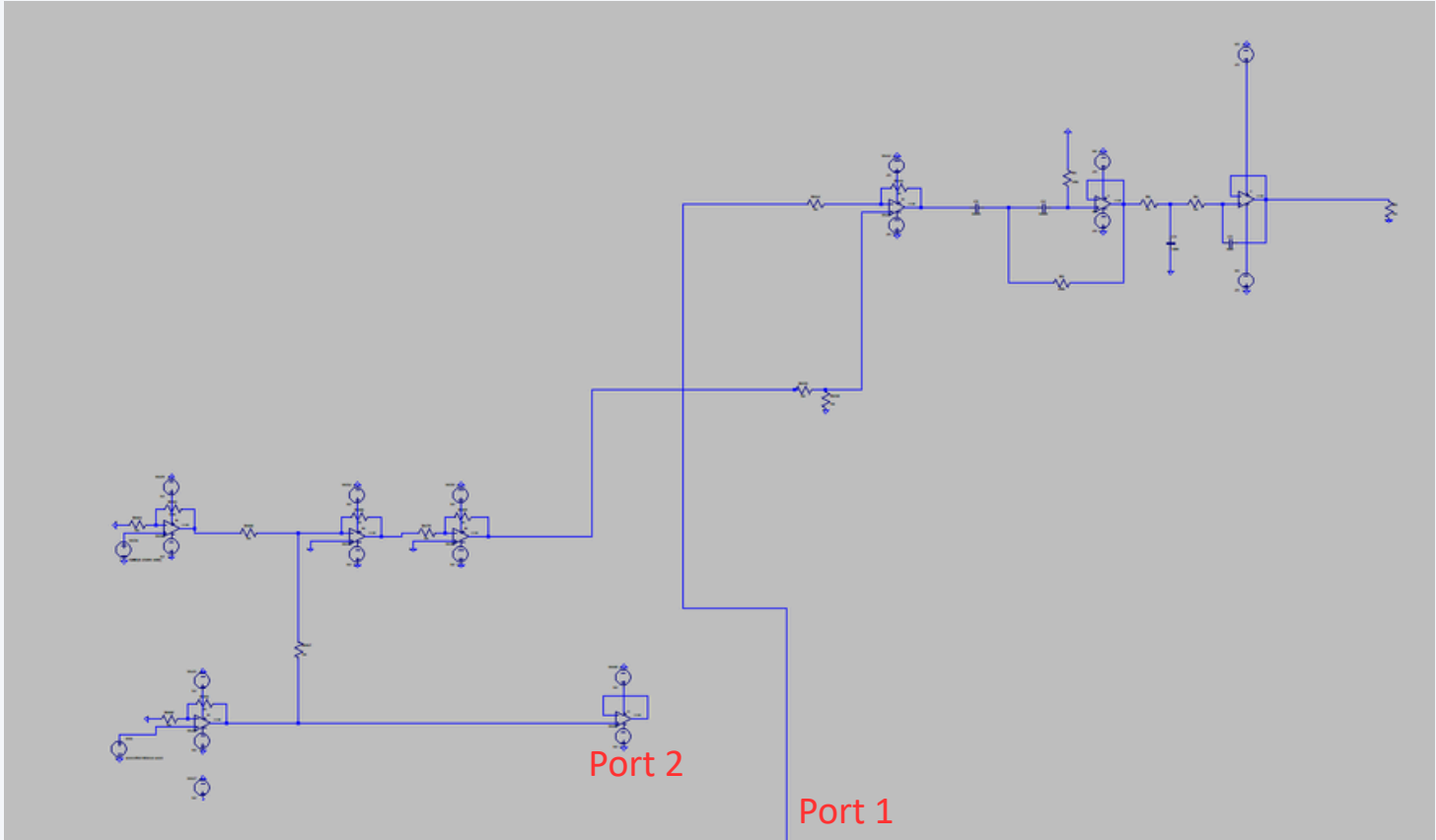
- SNR measures how strong the desired signal is compared to the unwanted noise.

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left[\left(\frac{A_{\text{signal}}}{A_{\text{noise}}} \right)^2 \right] = 20 \log_{10} \left(\frac{A_{\text{signal}}}{A_{\text{noise}}} \right) = A_{\text{signal,dB}} - A_{\text{noise,dB}}.$$

- A higher Signal-to-Noise Ratio (SNR) indicates better audio quality because the desired signal is much stronger than the unwanted noise.
- A lower Signal-to-Noise Ratio (SNR) indicates poorer audio quality because the noise becomes more noticeable compared to the desired signal.

SNR improvement = 5.329dB

02 — Hardware Implementation





Thank You.....



TRANSFER FUNCTION

$f_c = 1 / (2 \times \pi \times R \times C)$

COMPONENT VALUES

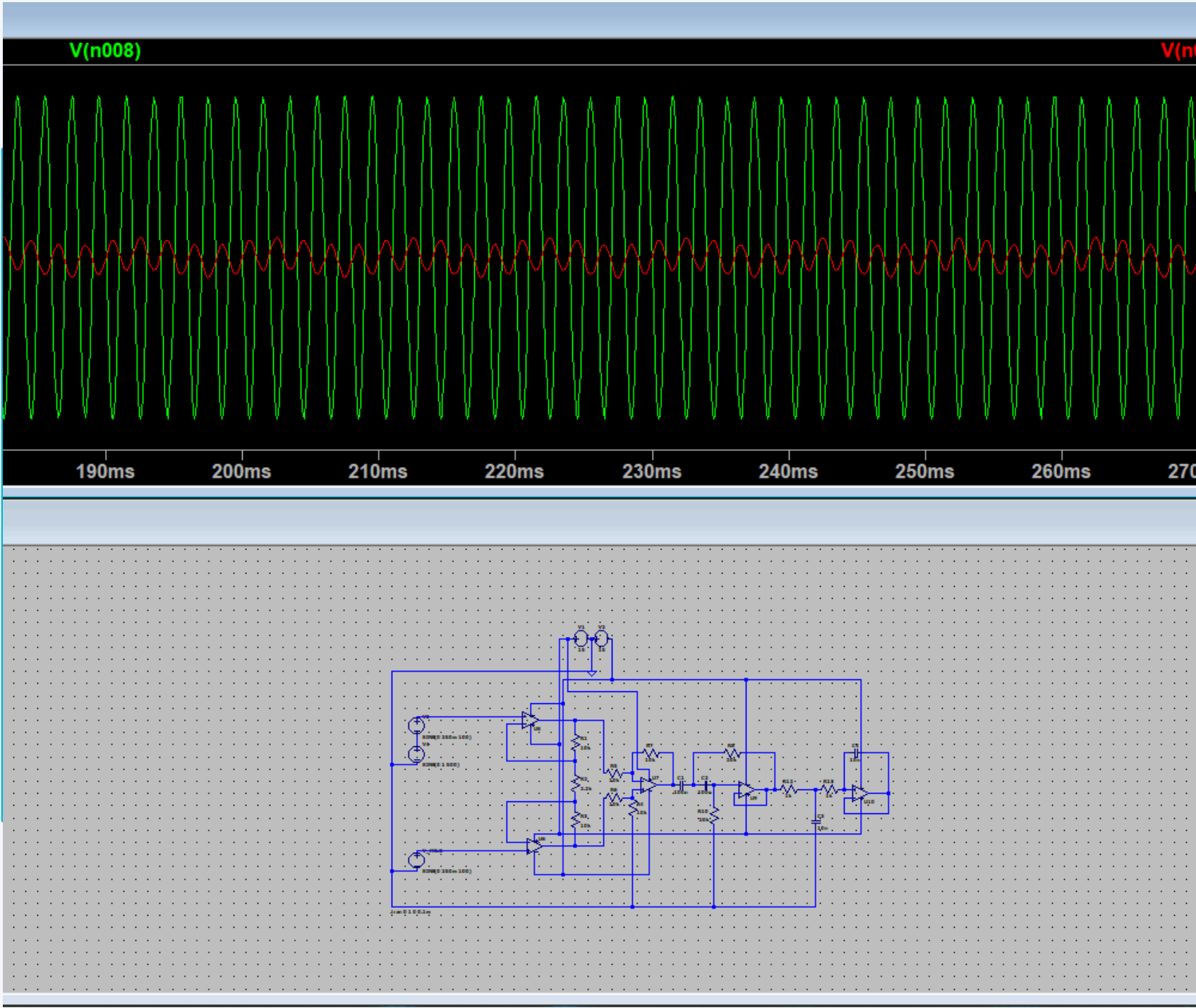
Stage	Topology	R	C	f _c (calc.)
HPF	2nd-Order Sallen-Key	22 kOhm	100 nF	72.3 Hz
LPF	2nd-Order Sallen-Key	1 kOhm	10 nF	15.92 kHz

Butterworth Response

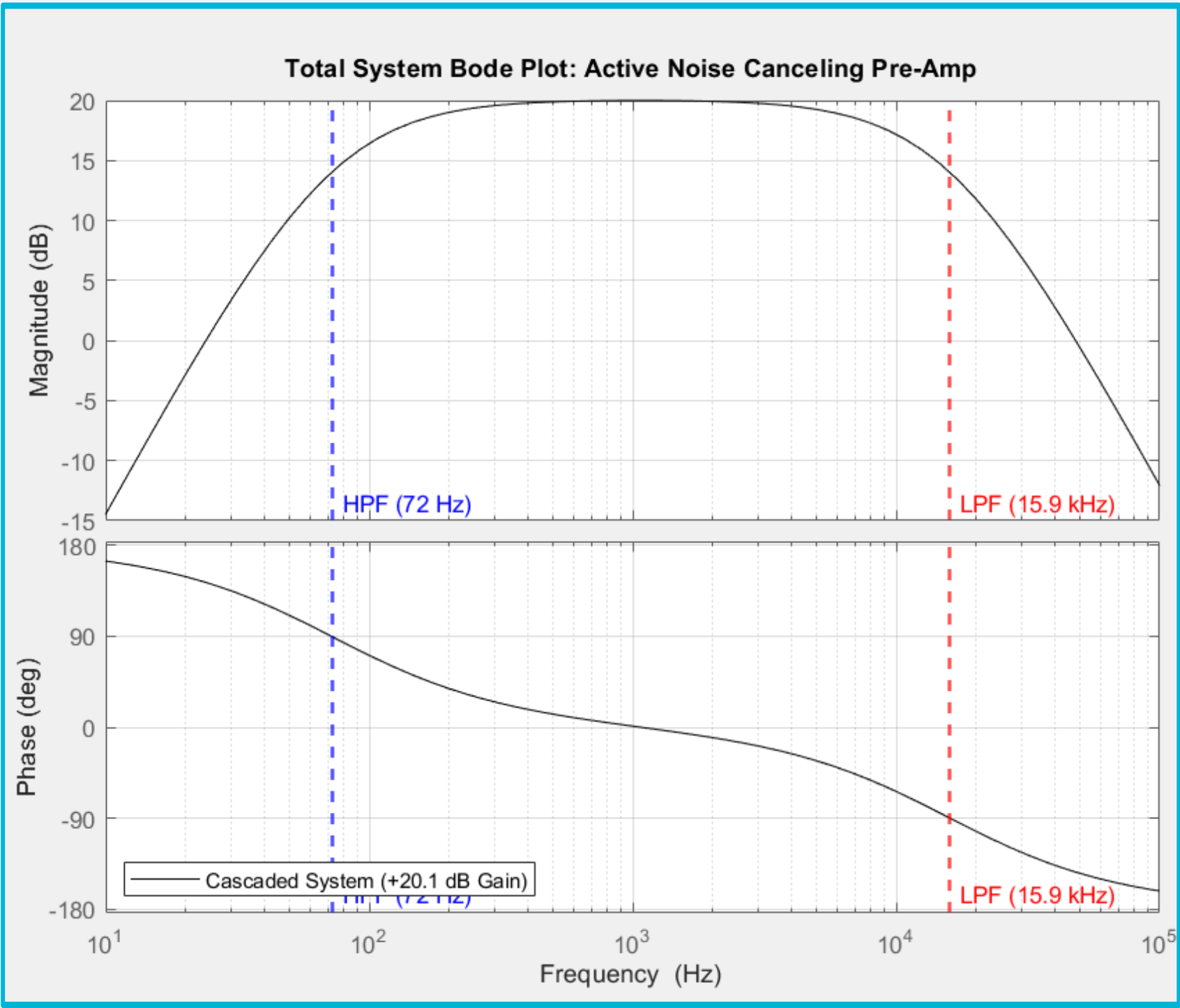
- Unity-Gain (Voltage-Follower) Sallen-Key topology
- Q = 1/sqrt(2) = 0.707 -> maximally flat passband (Butterworth)
- Component ratio R1=R2, C1=C2 satisfies Q = 0.707 exactly
- No resonant peaking — preserves voice-band frequency integrity
- Roll-off: -40 dB/decade beyond cutoff (2nd-order)
- Input impedance: high (op-amp non-inverting input)
- Gain sensitivity: S(Q) = 0.5 — low sensitivity to component spread

Verification: HPF -> f_c = 1/(2π × 22000 × 100e-9) = 72.3 Hz [OK] LPF -> f_c = 1/(2π × 1000 × 10e-9) = 15,915 Hz [OK]

05 — System Analysis & Simulation Results



LTspice — Transient Simulation



MATLAB — Frequency-Domain Analysis

Note: Practical CMRR limited by resistor matching tolerance (1% -> ~46 dB; 0.1% -> ~66 dB). Theoretical limit assumes ideal component matching.